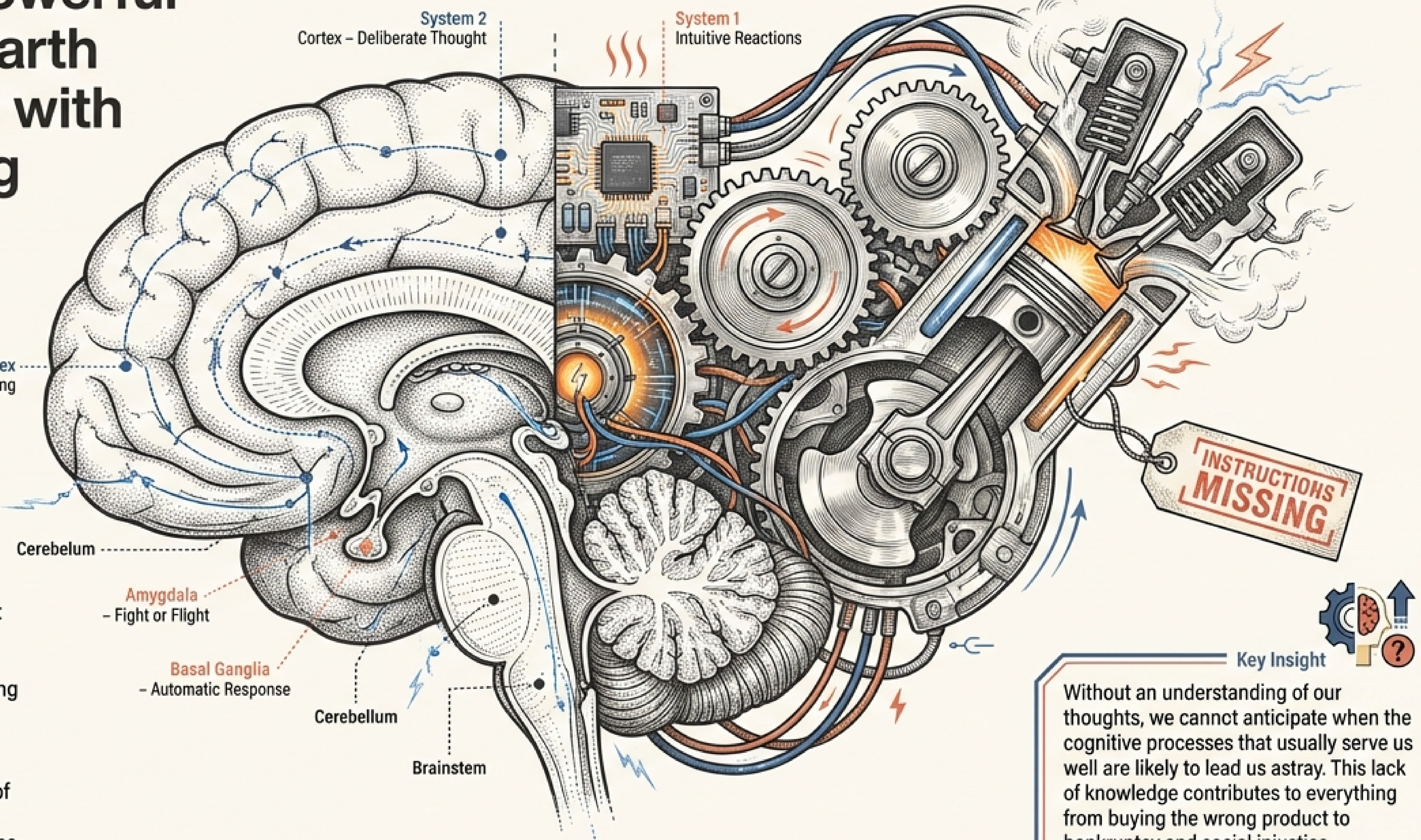
The most powerful engine on earth didn't come with an operating manual.



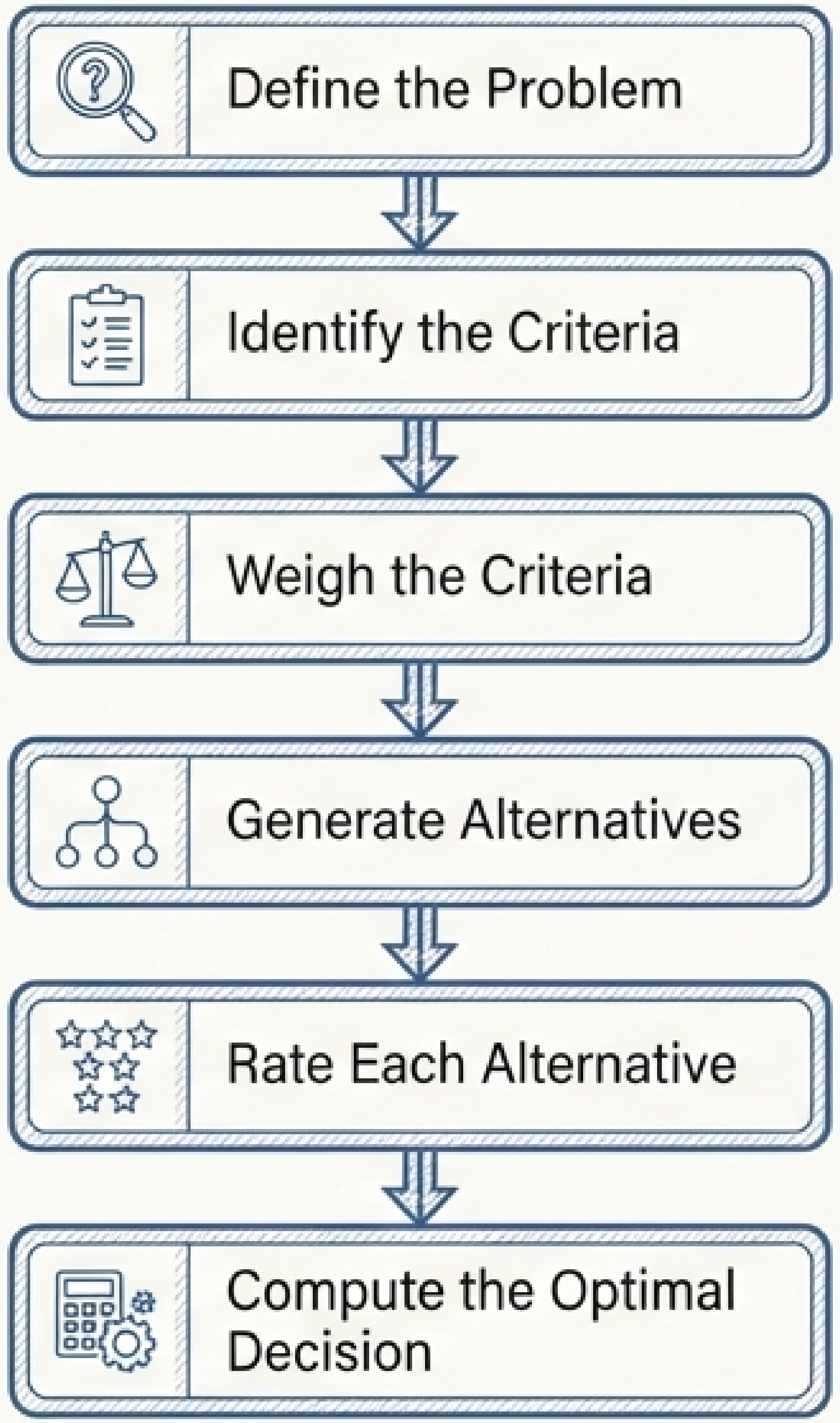
The human mind packs spectacular power into a modest three-pound mass. We accomplish sophisticated tasks—recognizing faces, catching balls—that baffle the most powerful computers.

Yet, we remain largely unaware of how our minds actually work. Introspection offers poor guidance.

Key Insight

Without an understanding of our thoughts, we cannct anticipate when the cognitive processes that usually serve us well are likely to lead us astray. This lack of knowledge contributes to everything from buying the wrong product to bankruptcy and social injustice.

The Myth of the Rational Process

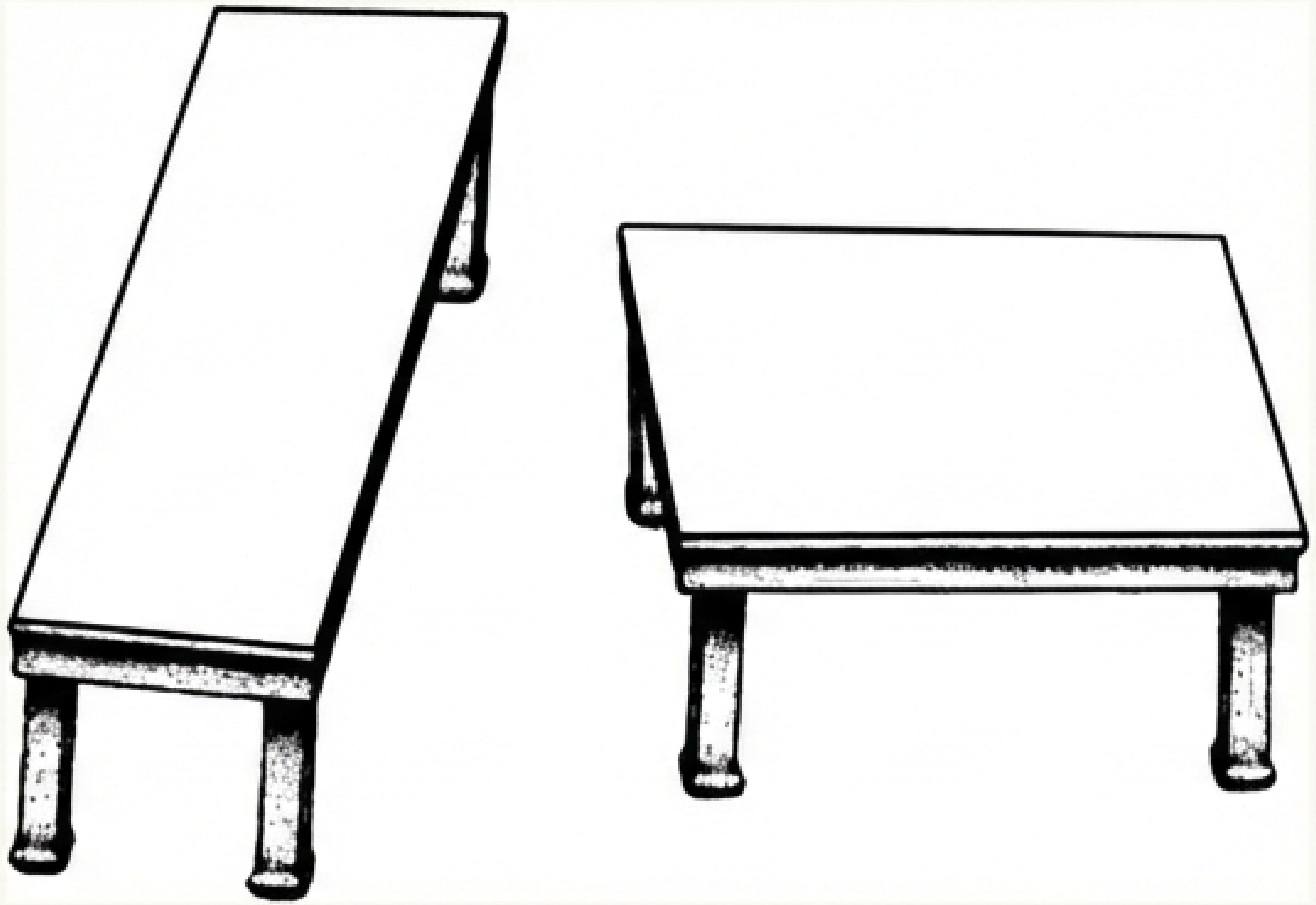


Do we actually do this?

In a perfect world, every managerial decision would follow these six precise steps to maximize value. However, research suggests that even the brightest people—those with the highest test scores—rarely employ these steps optimally.

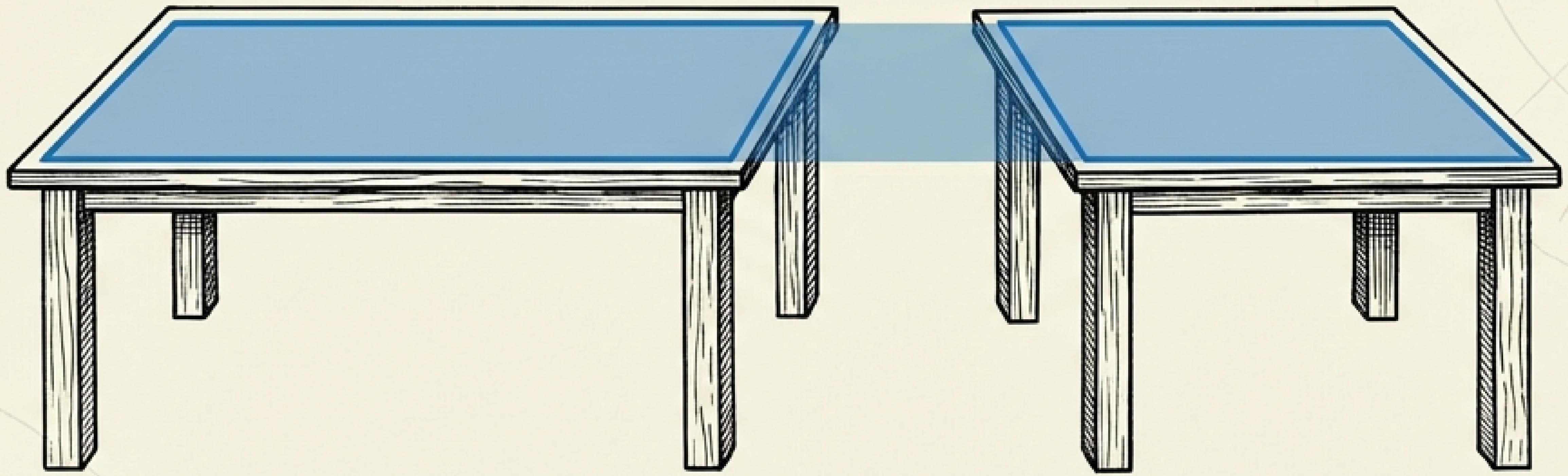
A Test of Your Intuition

Look at the two tables below.



Question: Which table is longer and thinner? Which one is more of a square? Take a moment. Trust your eyes. Commit to an answer before turning the page.

Your intuition just failed you.



Like most people, you likely saw the table on the right as a square and the left as long and skinny. In reality, they are identical.

The Lesson: This is not an eyesight problem; it is a processing problem. If you cannot trust your eyes to judge a simple drawing, why should you trust your gut to make complex managerial decisions?

Key Term: BIASES. These are systematic errors that occur when our intuitive system fails to adjust to reality.

The Two Engines Running Your Mind

System 1 (Intuitive)

- Fast & Automatic
- Effortless & Implicit
- Emotional
- Usage: Making most daily decisions (interpreting language, habits).
- Risk: Busy executives default here.

System 2 (Logical)

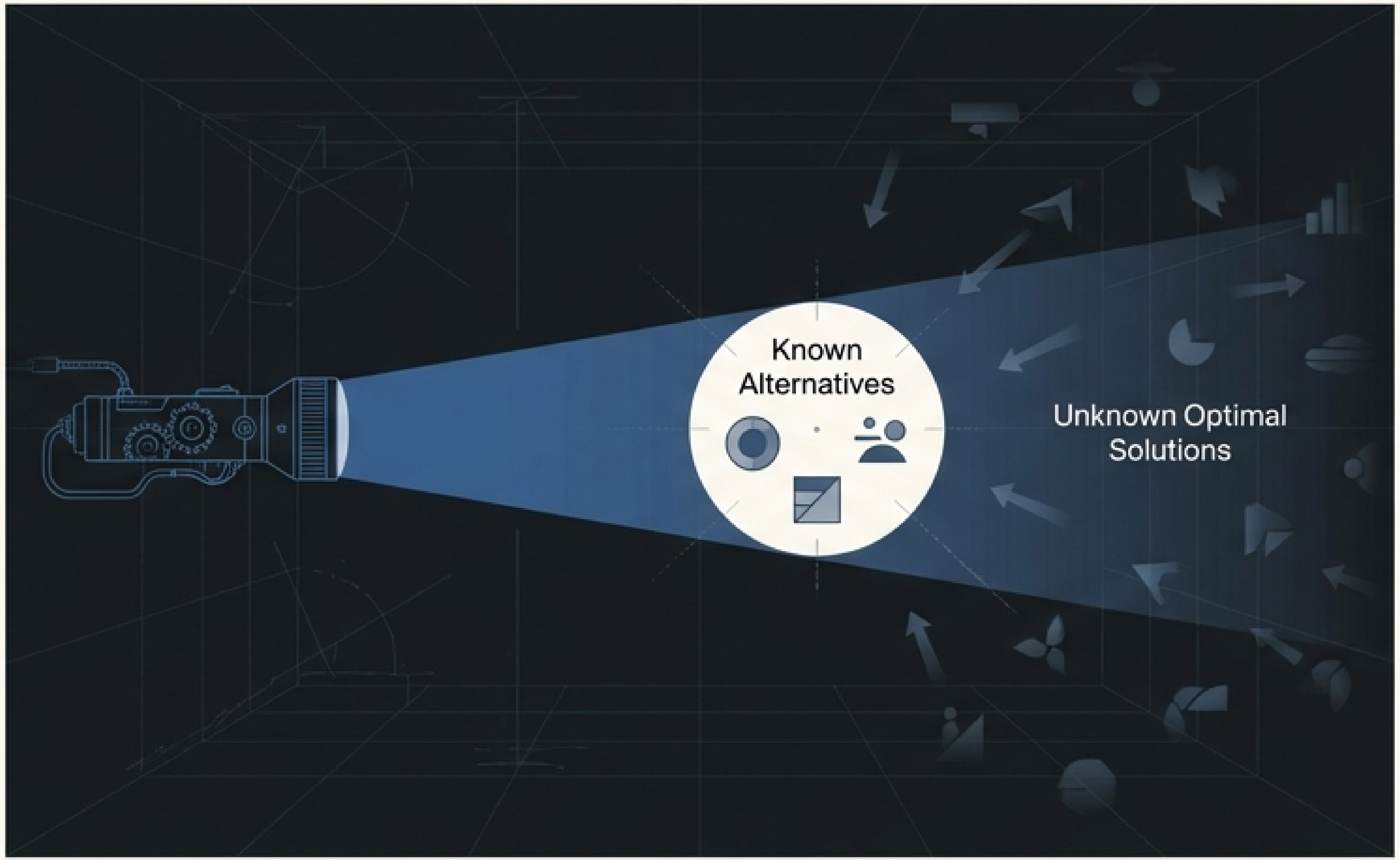
- Slow & Conscious
- Effortful & Explicit
- Logical
- Usage: Required for complex, novel decisions.
- Risk: Often neglected due to mental laziness.

System 2 logic should influence our most important decisions, yet we often default to System 1 because we have "too much on our minds."

We do not optimize. We “Satisfice.”

The rational model is Prescriptive (what we *should* do).

The reality is Descriptive (what we *actually* do).



The Theory

Nobel Laureate Herbert Simon introduced “Bounded Rationality.”

We lack the time, information, and brainpower to be perfectly rational.

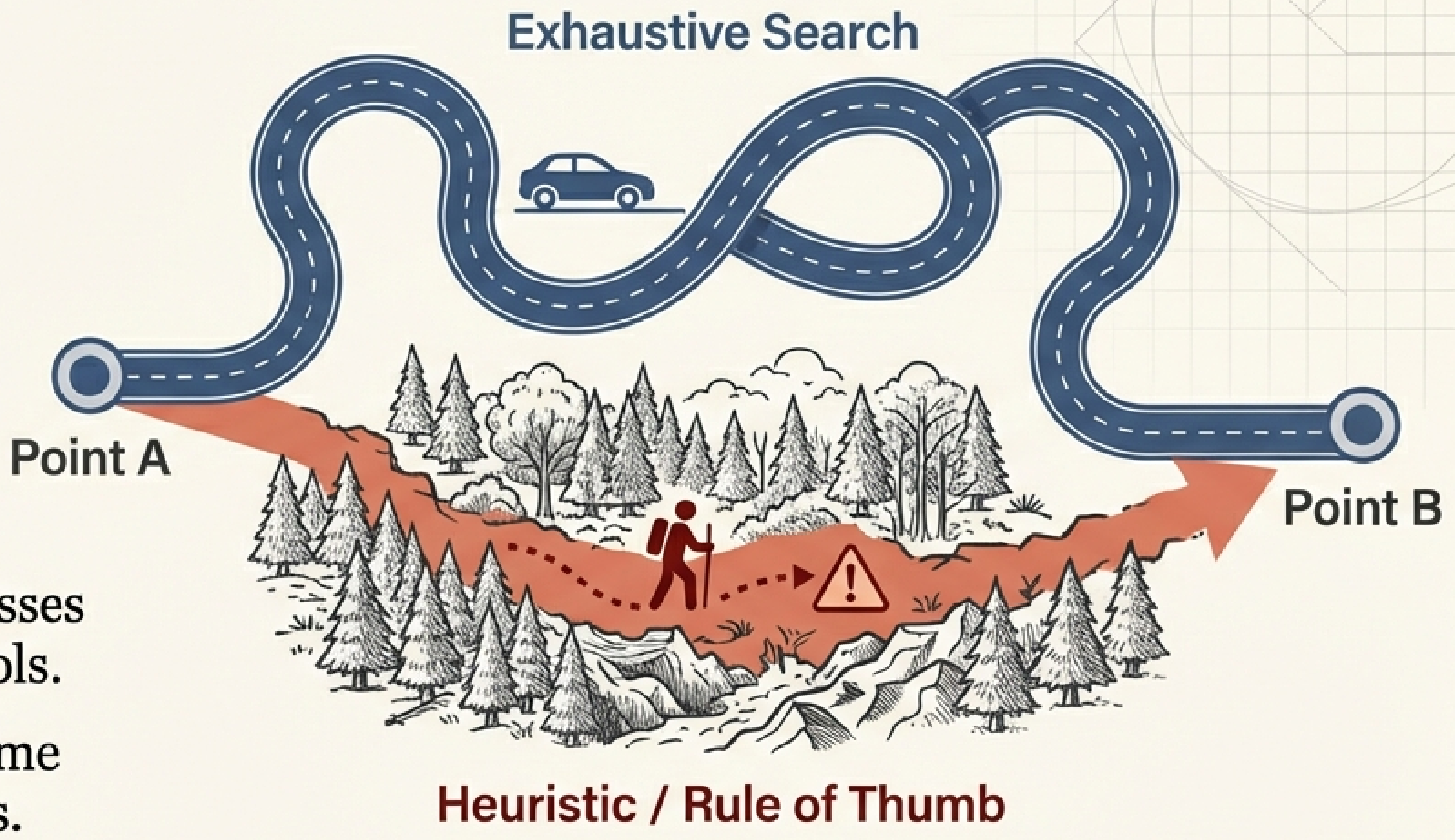
Instead of finding the BEST solution, we search only until we find that is “good enough.”

“SATISFICE = SATISFY + SUFFICE”

The Double-Edged Sword of Heuristics

Case Study: Marla Bannon

Marla needs a marketing MBA for her startup. To save time, she limits her search to only the top 6 management schools.

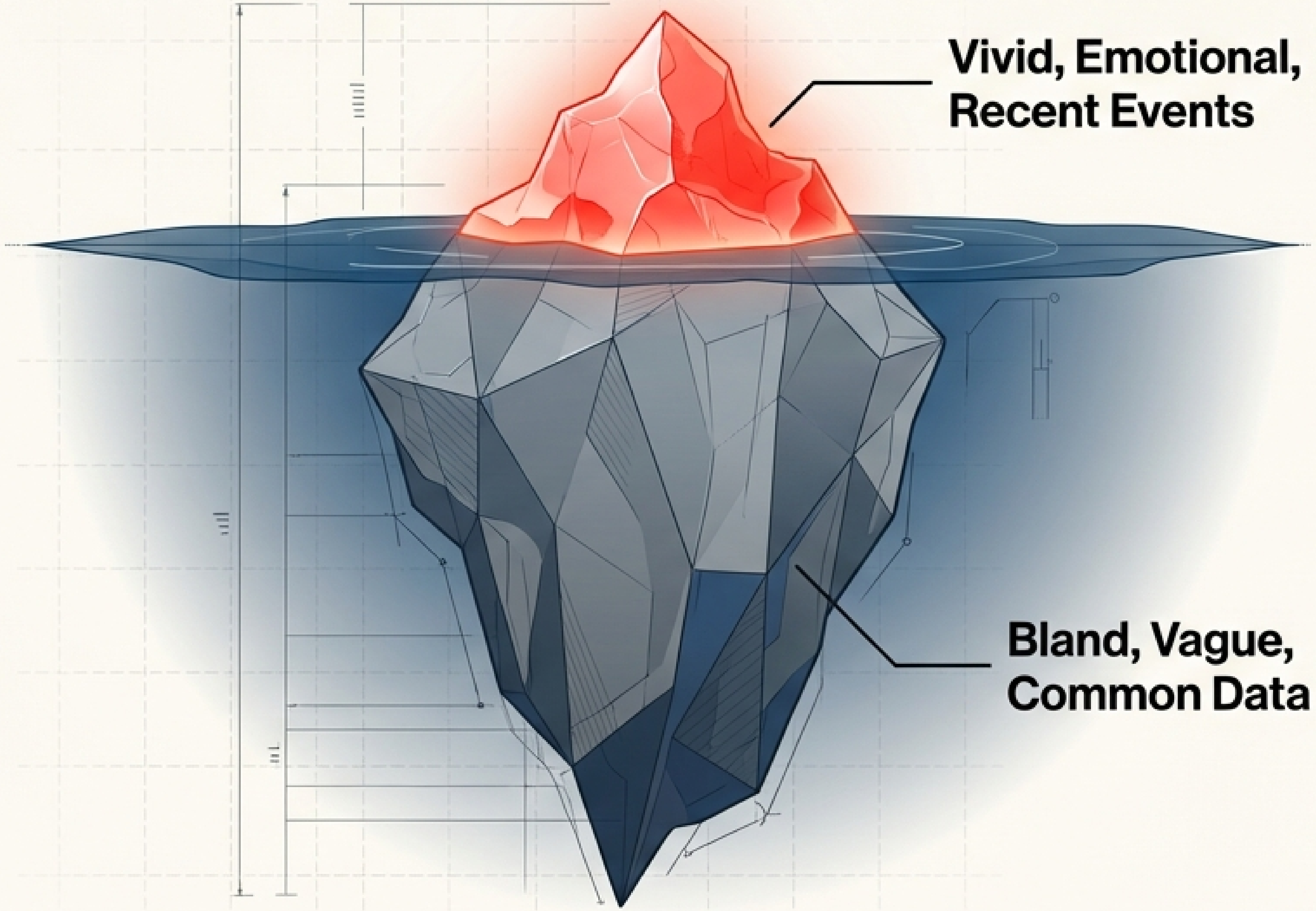


Analysis

- Rational View (Deficient): She misses talented candidates at other schools.
- Economic View (Efficient): The time saved outweighs the potential loss.

Heuristics are mental shortcuts. They help us cope with complexity, but we are often unaware we are using them.

Heuristic 1: Availability

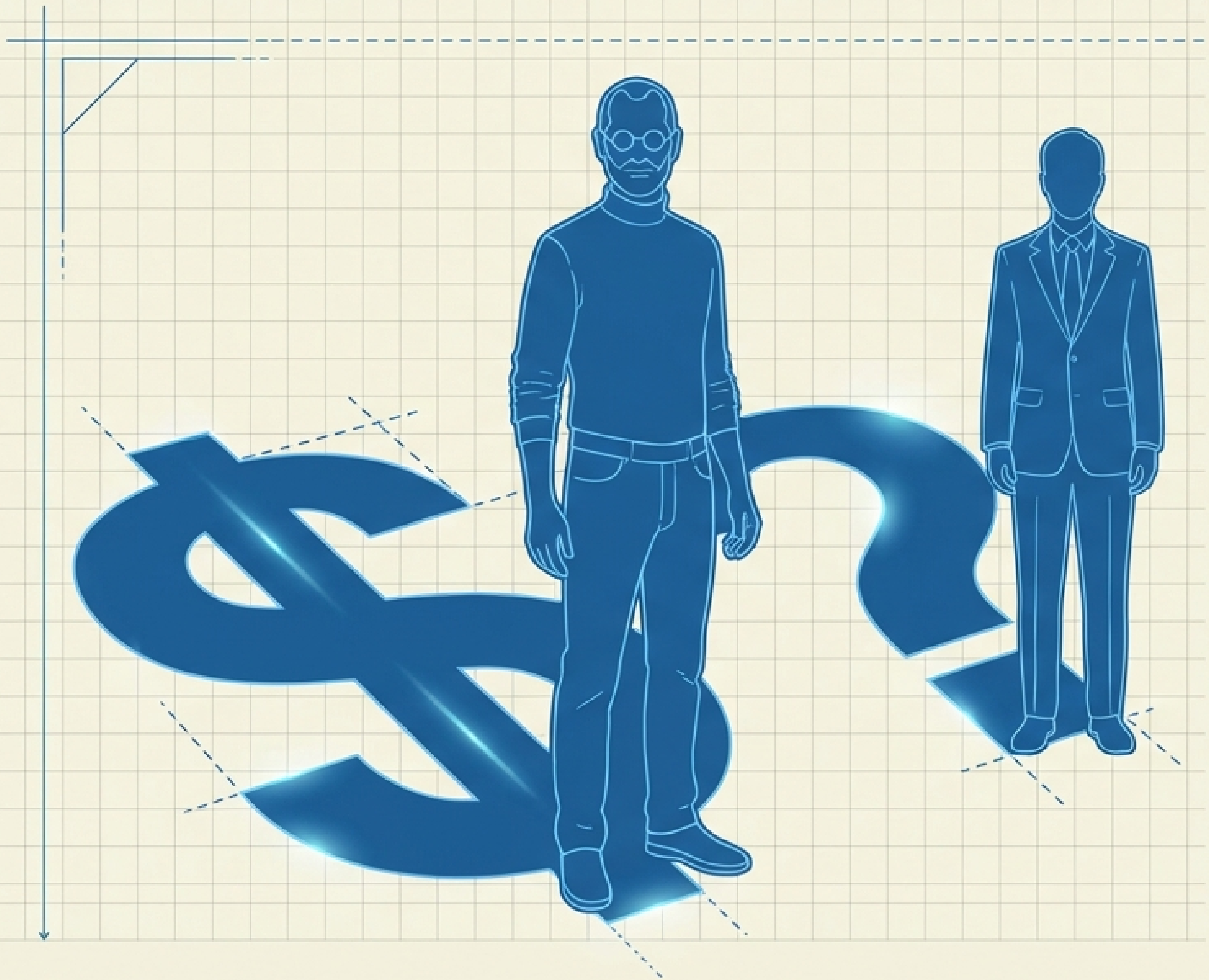


The Rule:
We assess the probability of an event by how easily instances of it come to mind.

The Bias:
Vivid memories (plane crashes, shark attacks, a nearby employee's mistake) feel more frequent than they statistically are.

Managerial Example:
Performance Reviews: A manager judges a nearby employee more harshly because their errors are physically visible, while ignoring the errors of a remote employee.

Heuristic 2: Representativeness



The Rule:
We predict success based on how much someone resembles a stereotype of success we already know.

The Bias:
Insensitivity to base rates. We ignore the statistics of failure because the story fits.

The "Bezos" Trap:
A venture capitalist funds an entrepreneur who *looks* like a famous founder, ignoring the actual quality of the business plan.

Historical Note:
This heuristic delayed the germ theory of disease. People couldn't believe a microscopic bug (unrepresentative) could cause a massive plague (huge consequence).

Heuristic 3: Confirmation

Is marijuana use related to delinquency?

To answer this, your brain likely tries to recall marijuana users you know and checks if they are delinquents.

STOP. You are testing your hypothesis only by looking for evidence that supports it. You are ignoring half the data.

The Data You Ignored

Marijuana Users / Delinquents We look here (Present/Present).	**Marijuana Users / Non-Delinquents** We ignore this.
Non-Users / Delinquents We ignore this.	**Non-Users / Non-Delinquents** We ignore this.

The Confirmation Trap: Our intuitive mind focuses only on the cell where the variable is present. A rational assessment requires analyzing all four groups to calculate the real correlation.

Heuristic 4: Affect

The Rule: Most judgments follow an emotional evaluation that occurs ***before*** logic kicks in.

The Evidence:

- Stock prices go up on sunny days (optimism induced by weather).
- Hiring decisions are influenced by whether a candidate reminds a manager of a disliked ex-spouse.
- Juries award penalties based on “outrage” rather than a logical assessment of harm.

The Mother of All Biases: Overconfidence

Among all the flaws in our judgment, Overconfidence is the most potent, pervasive, and pernicious.

“It is the most robust finding in the psychology of judgment.” — DeBondt & Thaler, 1995

The Cost: Wars, bubbles, bankruptcies, and lawsuits.

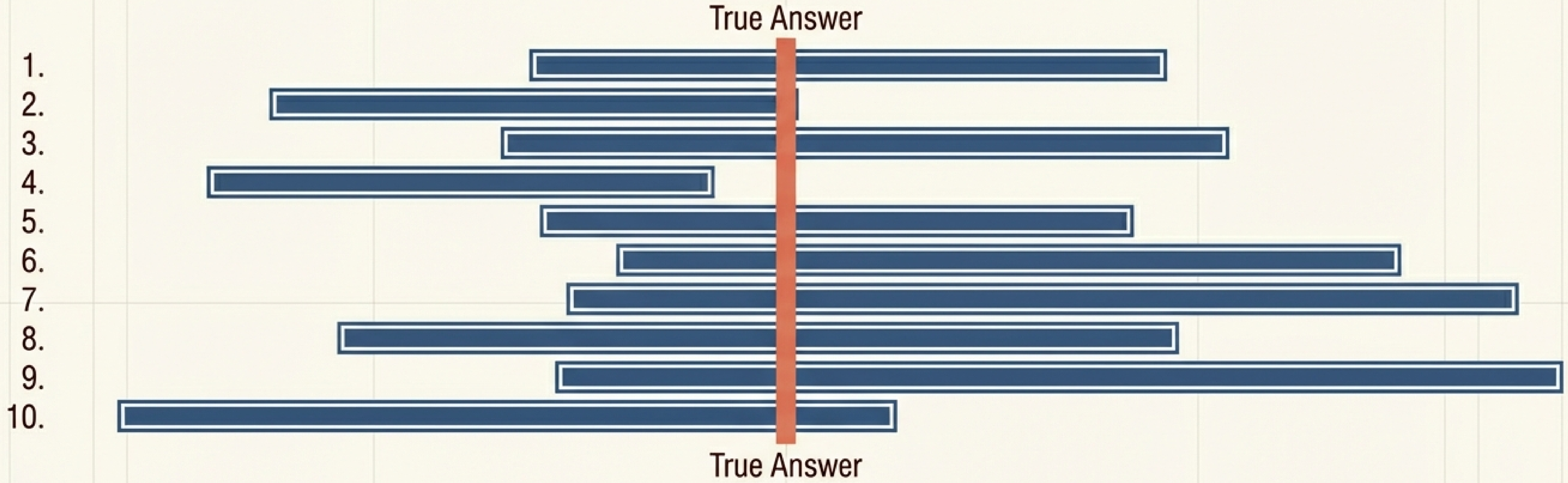
A Test of Confidence

For the items below, **do not** guess the exact number. Provide a RANGE (Low Estimate to High Estimate) so that you are 98% Confident the true answer lies within your range.

1. Wal-Mart's 2010 Revenue:		to		
2. Google's 2010 Revenue:		to		
3. World Population (Jan 2012):		to		
4. 2010 U.S. GDP:		to		
5. Population of China (Dec 2011):		to		
6. Rank of McDonald's in Fortune 500 (2010):		to		
7. Rank of General Electric in Fortune 500 (2010):		to		
8. Deaths due to motor vehicle accidents (2008, worldwide):		to		
9. U.S. National Debt (Dec 2011):		to		
10. National Debt of Greece (Dec 2011):		to		

Write down your ranges.

Why You Likely Failed the Test

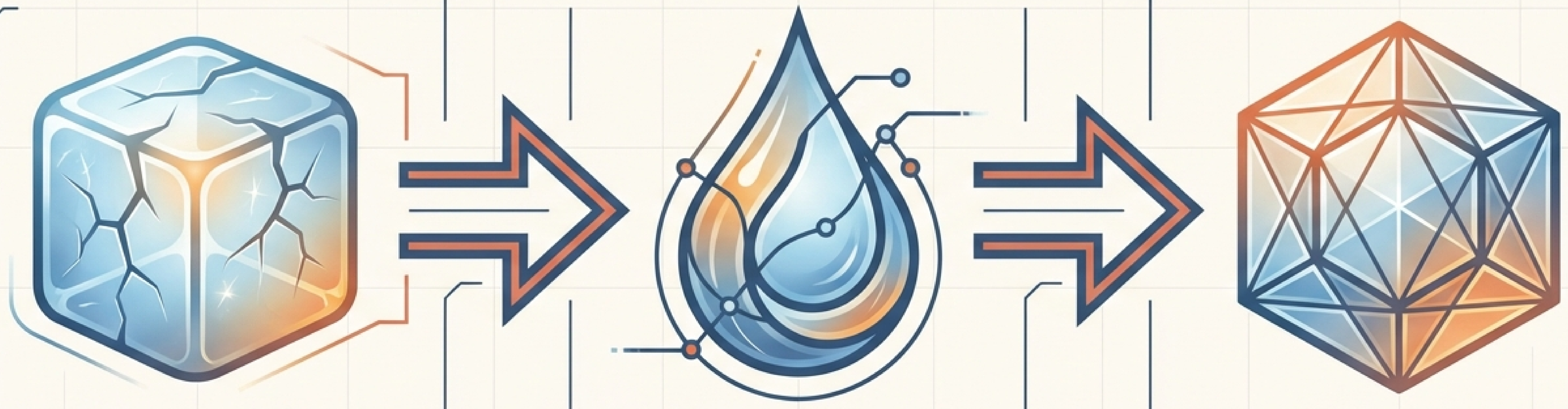


If you were truly 98% confident, you should only be wrong on 2% of the questions (0 or 1 question). Most people miss 4 to 7.

The Diagnosis: Overprecision. We anchor on a number we think is right and adjust slightly, but our adjustments are too tight. We are more confident in our knowledge than we should be.

Unfreeze. Change. Refreeze.

To improve judgment, simple awareness isn't enough. We must follow Lewin's Lewin's (1947) model of change to actively restructure how we decide.

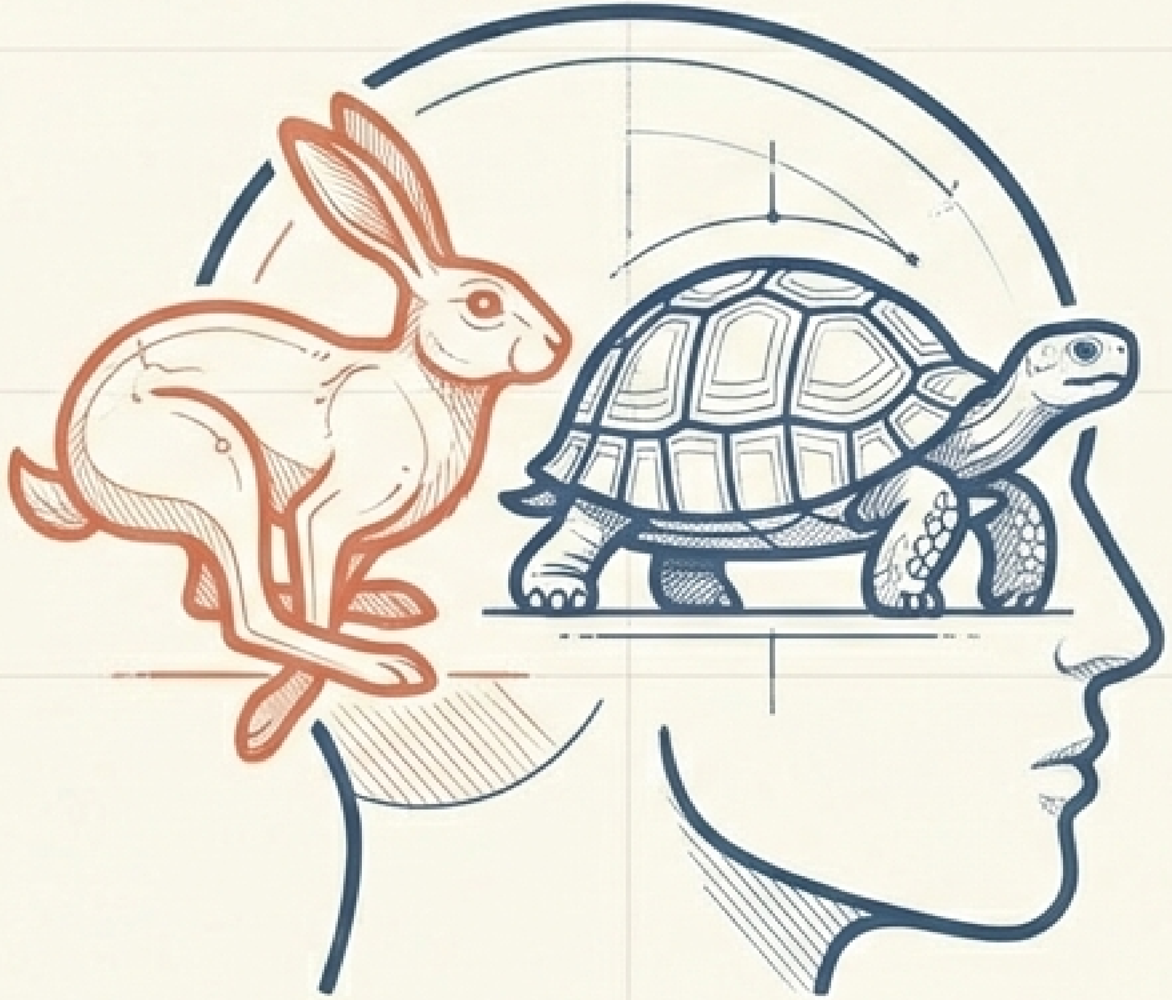


Unfreeze: Admit the flaw (Shepard Tables)

Change: Learn the Heuristics (System 1 vs System 2)

Refreeze: Make the new process permanent

The Goal is Not Perfection. It is Awareness.



We cannot simply ‘turn off’ System 1—nor should we. It helps us survive the day. The goal is to identify the specific high-stakes situations where our intuition is likely to fail, and consciously switch to System 2.

The difference between a good manager and a great one is the ability to spot the bias before the decision is made.